

Nervous tissues and nerve impulses

Nervous tissues contain cells which conduct 'messages' called **nerve impulses**. These cells are called **neurones**. The spaces between neurones are filled with cells called **neuroglia**, which support neurones, nourish them, and help them conduct impulses.

Neurones and nerve fibres

Like other cells neurones consist of a nucleus and cytoplasm. But unlike other cells the cytoplasm of a neurone extends outwards forming fine threads called **nerve fibres**. A nerve is a bundle of nerve fibres bound together by connective tissue (Fig. 3A). Nerve fibres can be over one metre in length, and between 0.004 mm and 0.02 mm thick. The thicker the fibre the faster impulses travel along it. The thickest fibres conduct impulses at 120 metres per second. Nerve fibres outside the brain and spinal cord are enclosed in a sheath of fatty insulating material called **myelin**. This material is contained in the cytoplasm of special cells which wrap themselves around the nerve fibres (Fig. 2C).

Nerve cells which conduct impulses from the sense organs (eyes, ears, etc.) to the brain and spinal cord are called **sensory neurones** (Figs. 1 and 2A). Inside the brain and spinal cord impulses pass from sensory neurones to **association neurones** which, in turn, pass them to **motor neurones** (Fig. 1). Motor neurones (Fig. 2B) conduct impulses out of the brain and spinal cord to the muscles and glands.

Nerve impulses

Light, sound, touch, and heat are called **stimuli** (*singular stimulus*) because they stimulate sense organs into sending impulses along nerves. An impulse can be described as a travelling wave of electrical and chemical changes. There is no such thing as a strong or a weak impulse, and there is no difference between impulses from eyes, ears, or other sense organs. All impulses are exactly alike. The only feature of impulses which varies is the number per second travelling along a fibre. A very strong stimulus, such as a loud noise, can cause up to 100 impulses per second to travel from the ears to the brain. Alternatively there are sounds too faint to stimulate the ears. Stimulation must reach what is called the **threshold level** of a sense organ before it begins sending out impulses.

Each neurone is connected with many other neurones making literally millions of interconnections within the nervous system. Neurones are not continuous with one another. There is a gap called a **synapse** where they meet (Fig. 3 B and C). Impulses arriving at a synapse cause the release of chemicals which diffuse across the gap and stimulate the production of impulses in the next neurone.

Each synapse is a barrier to nerve impulses, a barrier cannot be crossed unless impulses are arriving at a certain number per second. These millions of barriers ensure that weak stimuli are ignored, and prevent impulses in one fibre from running through nerve interconnections to stimulate all the neurones in the body.

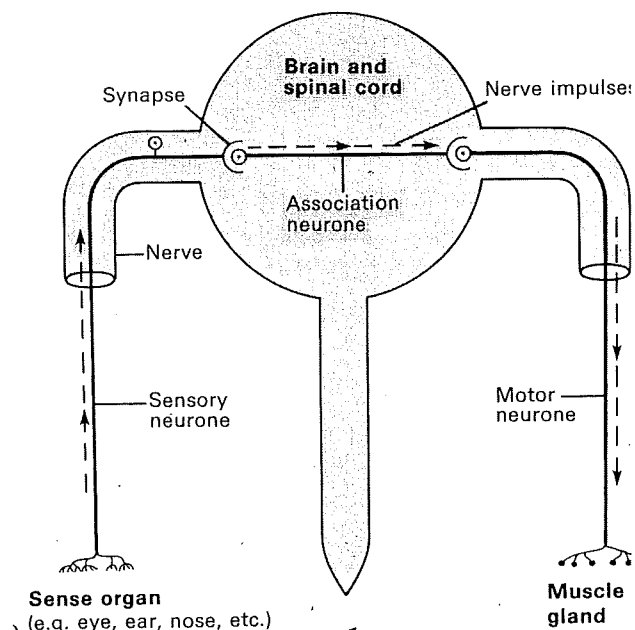
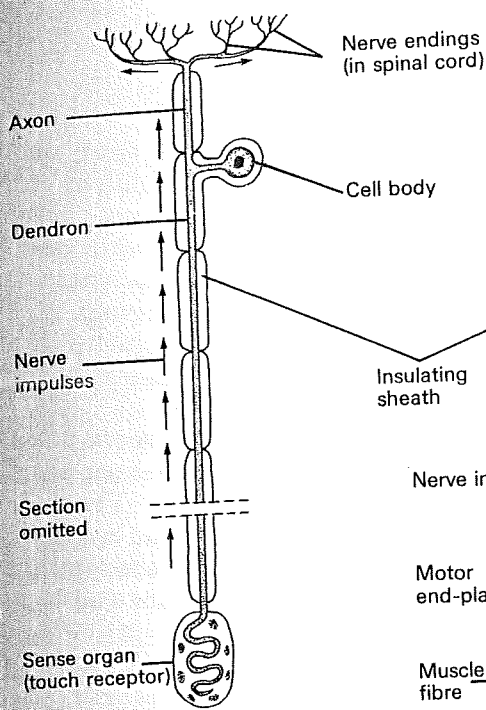
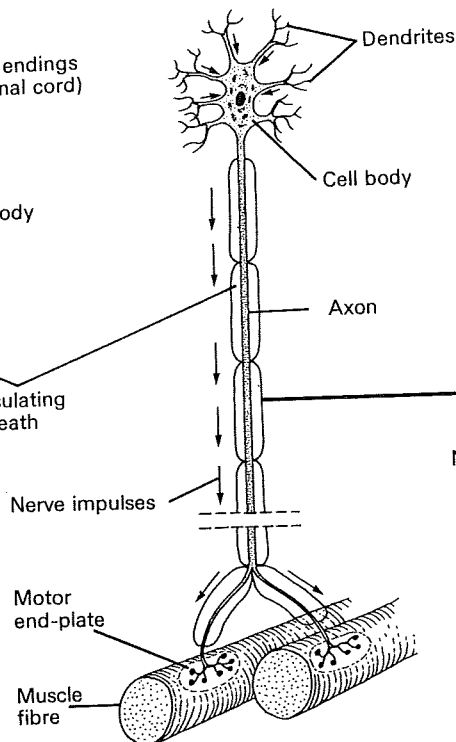


Fig. 1. Diagram of how nerve cells (neurones) are arranged in the nervous system. Sensory neurones carry impulses from sense organs to the brain or spinal cord. Association neurones carry impulses through the brain or spinal cord and pass them to motor neurones which carry them to muscles or glands.

A A typical sensory neurone



B A typical motor neurone



C Part of a nerve fibre (magnified)

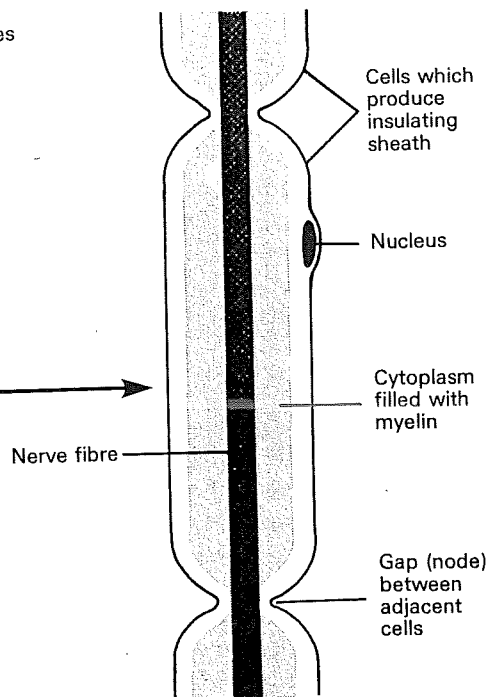
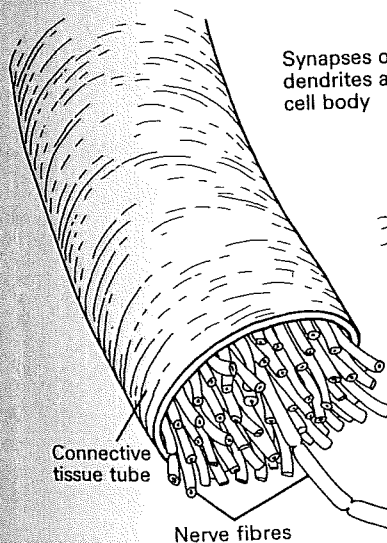


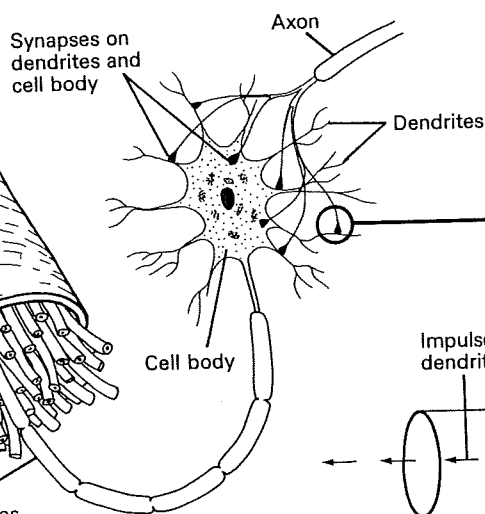
Fig. 2 A and B. Sensory and motor neurones. Neurones consist of a cell body and nerve fibres. Depending upon its length, a fibre which carries impulses towards the cell body is called a dendron (long) or dendrite (short). An axon carries impulses away from the cell body.

Fig. 2C. Nerve fibres outside the brain and spinal cord are covered by a sheath of fatty insulating material called myelin. The myelin is produced by cells which wrap themselves around the nerve fibres.

A Structure of a nerve



B Synapses



C Synapse (highly magnified)

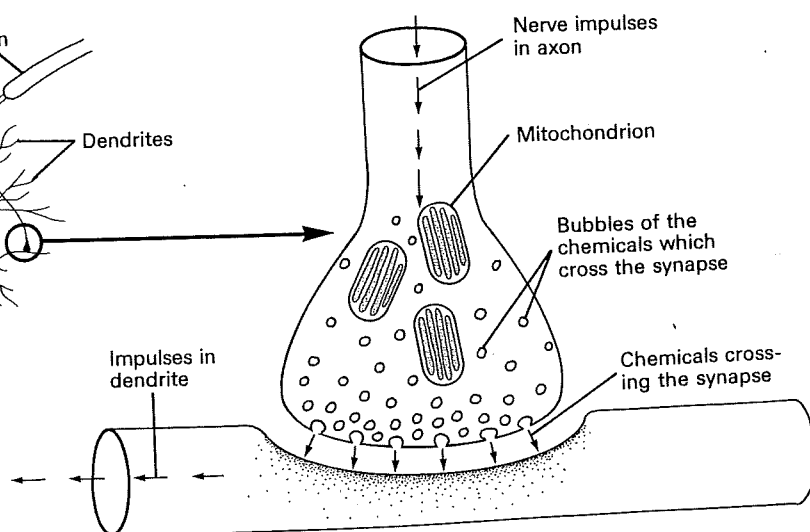


Fig. 3 A and B. A nerve is a bundle of nerve fibres in a tube of connective tissue.

Fig. 3C. There is a gap called a synapse where one nerve fibre meets another. Impulses arriving at a synapse release chemicals which cross the gap and stimulate the next neurone.

Reflexes

If your hand touches something very hot you pull it away quickly without having to think about what to do. If dust blows into your eyes tears are produced and blinking occurs automatically; and if a bright light shines in your eyes the pupils contract without conscious thought. All these are examples of **reflex actions**. In technical terms a reflex action is behaviour in which a stimulus produces an immediate and automatic response.

The nerve connections responsible for a typical reflex action are illustrated in Figure 1. These connections form what is called a **reflex arc**.

Control of reflexes

Figure 2 illustrates nerve fibres which convey impulses from a reflex arc up the spinal cord to the brain and back again. These nerve fibres enable the brain to be aware of reflex actions and, in some instances, to prevent them happening if they are inconvenient. To do this, the brain sends impulses down to a reflex arc to inhibit (stop) the reflex action taking place. It does this by preventing impulses

from travelling along the motor fibre of the arc. For example, provided the stimulus is not too strong, reflex response to pain can be prevented so that it is possible to hold on to something hot even though it is burning the fingers.

Nerve connections between reflex arcs at the brain allow **conditioned reflexes** to be established.

Conditioned reflexes

In 1902 the Russian scientist Ivan Pavlov discovered how to modify certain reflex actions of dogs. A hungry dog will respond to food by producing saliva. This is a normal reflex action. Pavlov discovered that if a bell is sounded immediately before the food is presented the dog will, after many repetitions, salivate in response to the bell alone. Salivation in response to food has been replaced by salivation in response to a bell. This is called **conditioning**. Figure 3 illustrates an example of human conditioning.

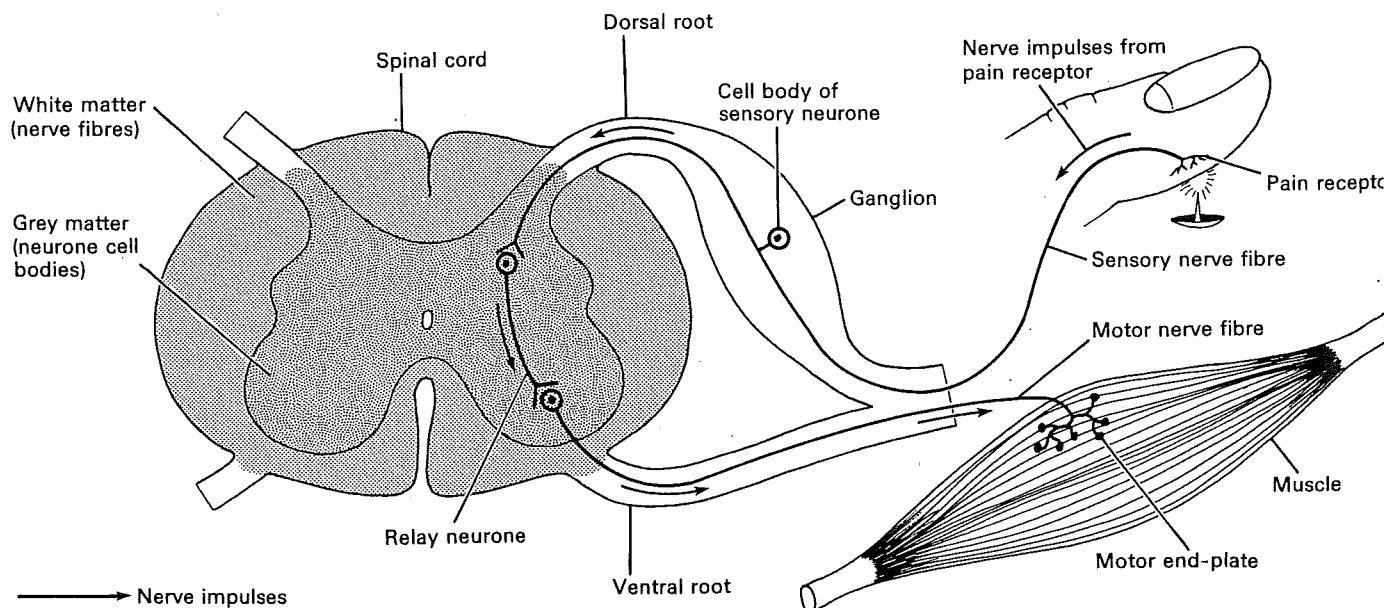


Fig. 1. A reflex action is a quick automatic response to a stimulus. This is a diagram of the nerve connections responsible for a typical reflex. Impulses from a sense organ (e.g. a pain receptor in a finger) travel along a sensory neurone into a dorsal root of the nerve cord. Here the impulses travel along a relay

neurone and out of the nerve cord along a motor neurone to the muscle which responds to the stimulus (in this case muscles which pull the finger from the stimulus). These nerve connections are an example of a reflex arc.

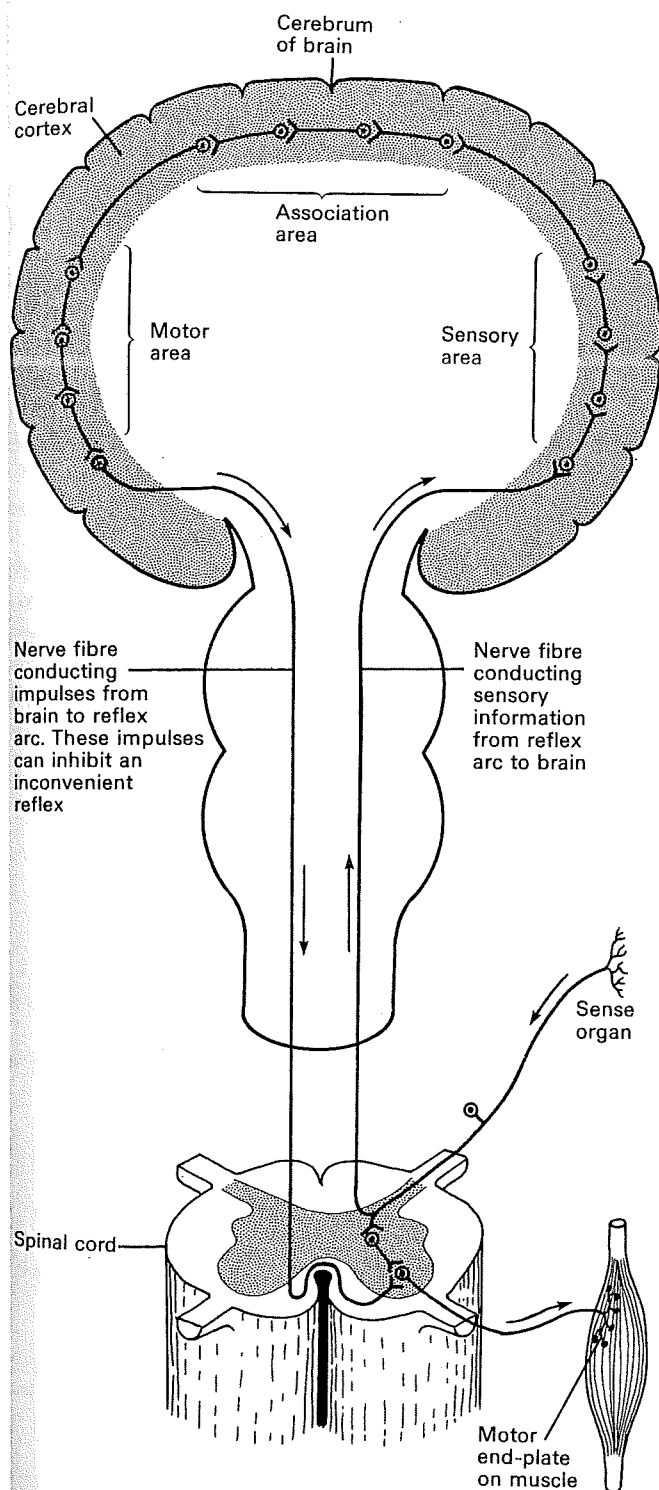
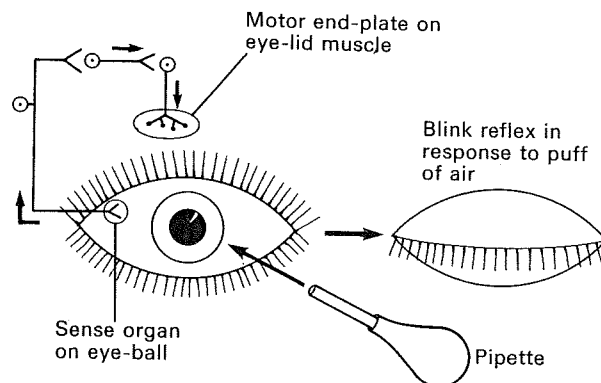
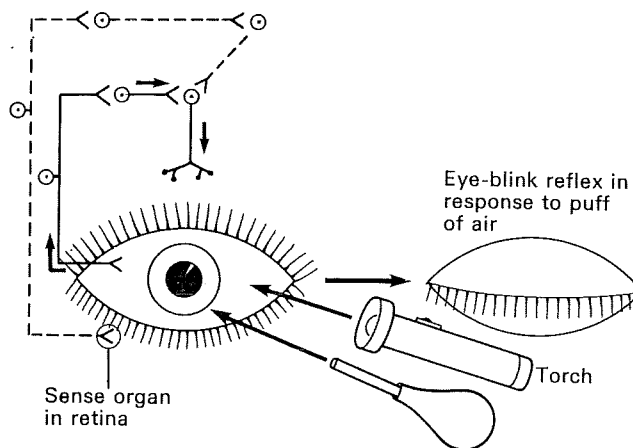


Fig. 2. How the brain can control reflexes. The brain is made aware of reflex actions by impulses travelling up fibres from the sensory neurone of a reflex arc. The brain can inhibit (stop) some reflexes by sending impulses down to the reflex arc.

A The eye-blink reflex occurs in response to a puff of air from a pipette



B The eye is subjected to a puff of air and to light several times. New nerve connections are established in the brain



C Eye-blink occurs in response to light alone

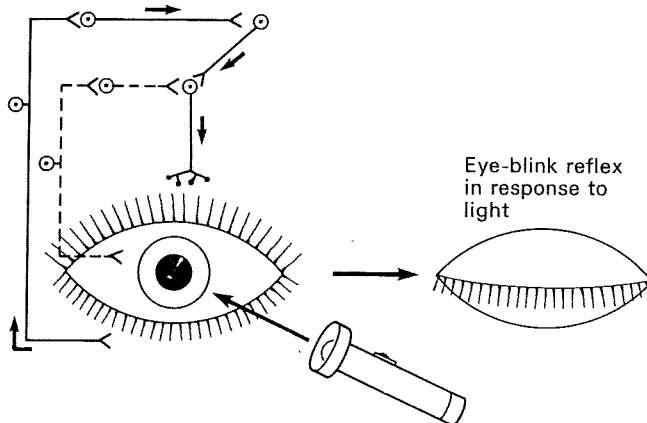


Fig. 3. An example of a conditioned reflex. Eyes normally blink in response to a puff of air and not light. But, after conditioning, they will blink in response to light alone.

Sense of balance

Many different sense organs play a part in helping us to maintain our balance. The eyes supply information about the position and movements of the body, and so do hundreds of stretch receptors in the muscles and joints. The parts of the inner ear other than the cochlea are also concerned with balance. The **semi-circular canals** detect changes in the direction of movements; and the **utricle** and **sacculus** detect changes in the speed of movements (acceleration and deceleration) and changes in the position of the body (whether it is upright or tilted).

Semi-circular canals

These are curved tubes surrounded by perilymph and filled with endolymph (Fig. 1). At one end of each semi-circular canal is a swelling called an **ampulla**. These contain sensory hair cells. The hairs of these cells are embedded in a cone of jelly called a **cupula** (Fig. 2A). When the head moves it causes endolymph in one or more of the canals to press against a cupula and bend it over (Fig. 4). This stretches the sensory hair cells which send impulses to the brain. The brain calculates the direction of the movement from the frequency of nerve impulses it receives from each ampulla.

Utricles and sacculles

These are spaces in the skull filled with endolymph (Figs. 1 and 2). There is a patch of sensory hair cells on the inner surface of each utricle and saccule (Fig. 2). The hairs of these cells are embedded in a mass of jelly containing tiny pieces of chalk called **otoliths**. Together, the hair cells, jelly, and otoliths are called a **sensory macula** (Figs. 2B and 3). The otoliths add considerably to the weight of the jelly. When the head is upright and still, the otoliths press down on the hair cells, but if the head tilts or is suddenly accelerated or decelerated, the otoliths pull against the hair cells causing them to send nerve impulses to the brain (Fig. 3). The brain calculates the angle of tilt or change in speed from the frequency of impulses which it receives from each utricle and saccule.

When the body is still few impulses reach the brain from the ampullas and maculas, but when the body is tilted and swayed as happens on a ship in rough seas, the brain receives a barrage of impulses from these organs. This over-stimulation may upset other areas of the brain and cause sea sickness.

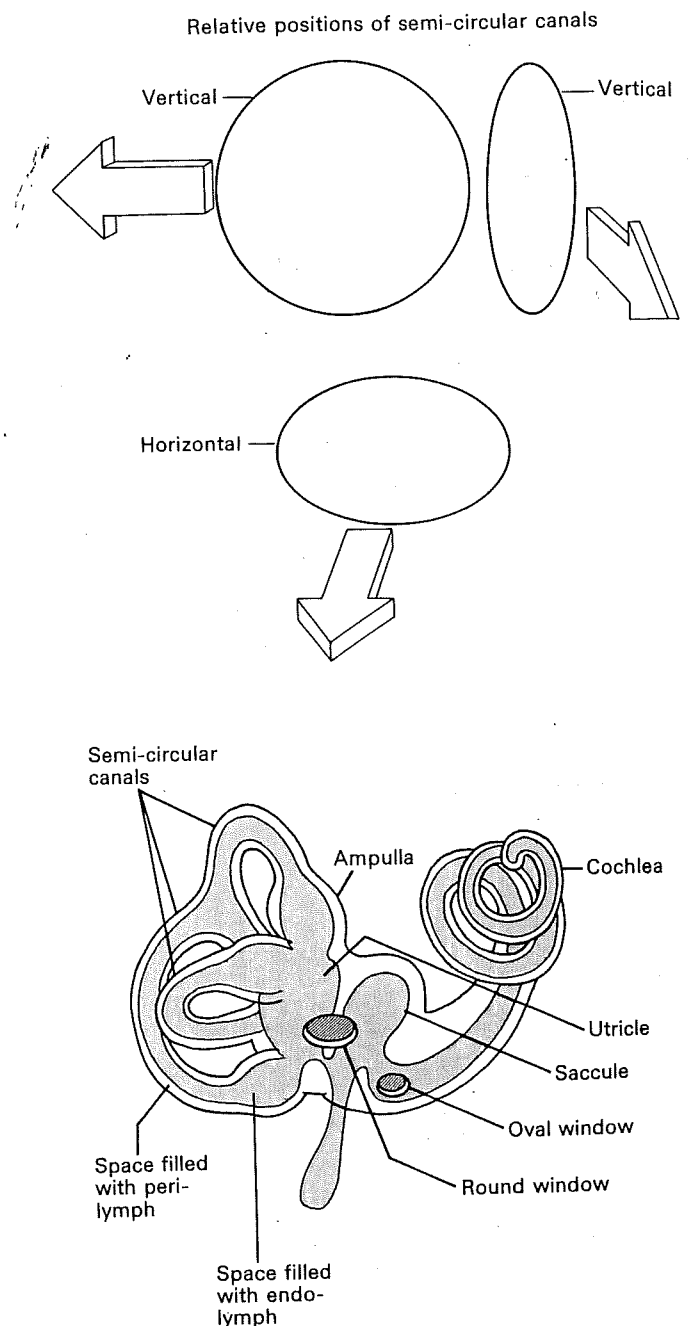


Fig. 1. The inner ear. This is also called the labyrinth because it consists of a complex series of passageways embedded in bone. The passages concerned with balance are the semi-circular canals, the utricle, and the sacculus. Semi-circular canals detect changes in the direction of movements. Utricles and sacculles detect acceleration, deceleration, and tilting of the head.

Structure of an ampulla

B Structure of a macula

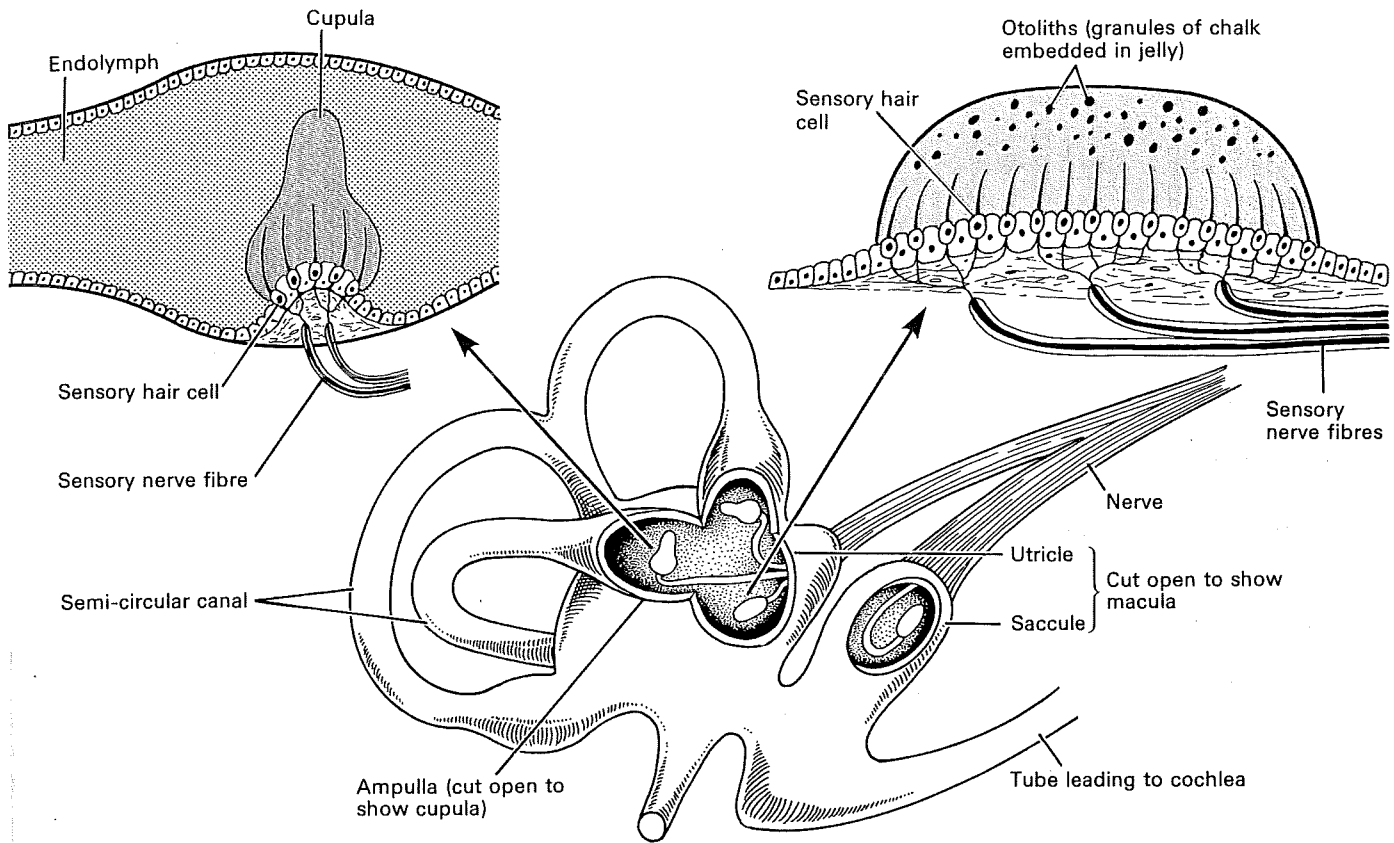
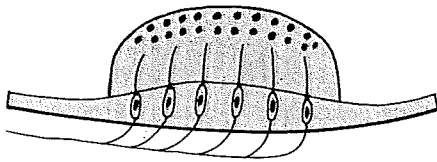


Fig. 2. An ampulla and utricle cut open to show the position of the cupula and macula.

Head level



Head tilted

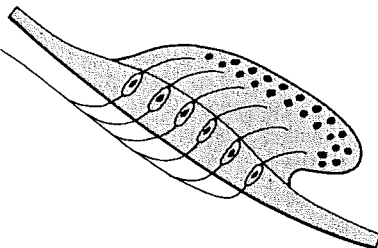


Fig. 3. How a macula works. When the head is level the otoliths press downwards. When the head tilts, speeds up, or slows down, the otoliths pull against the hair cells which send impulses to the brain.

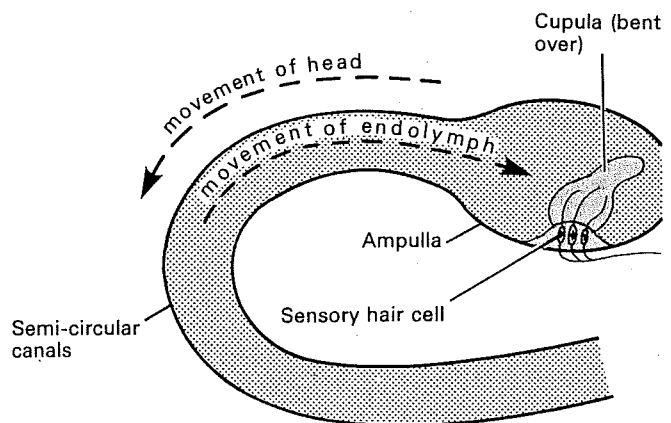


Fig. 4. How semi-circular canals work. When the head moves one way endolymph is 'left behind' and apparently flows the other way, bending the cupula. This stimulates the hair cells which send impulses to the brain.